

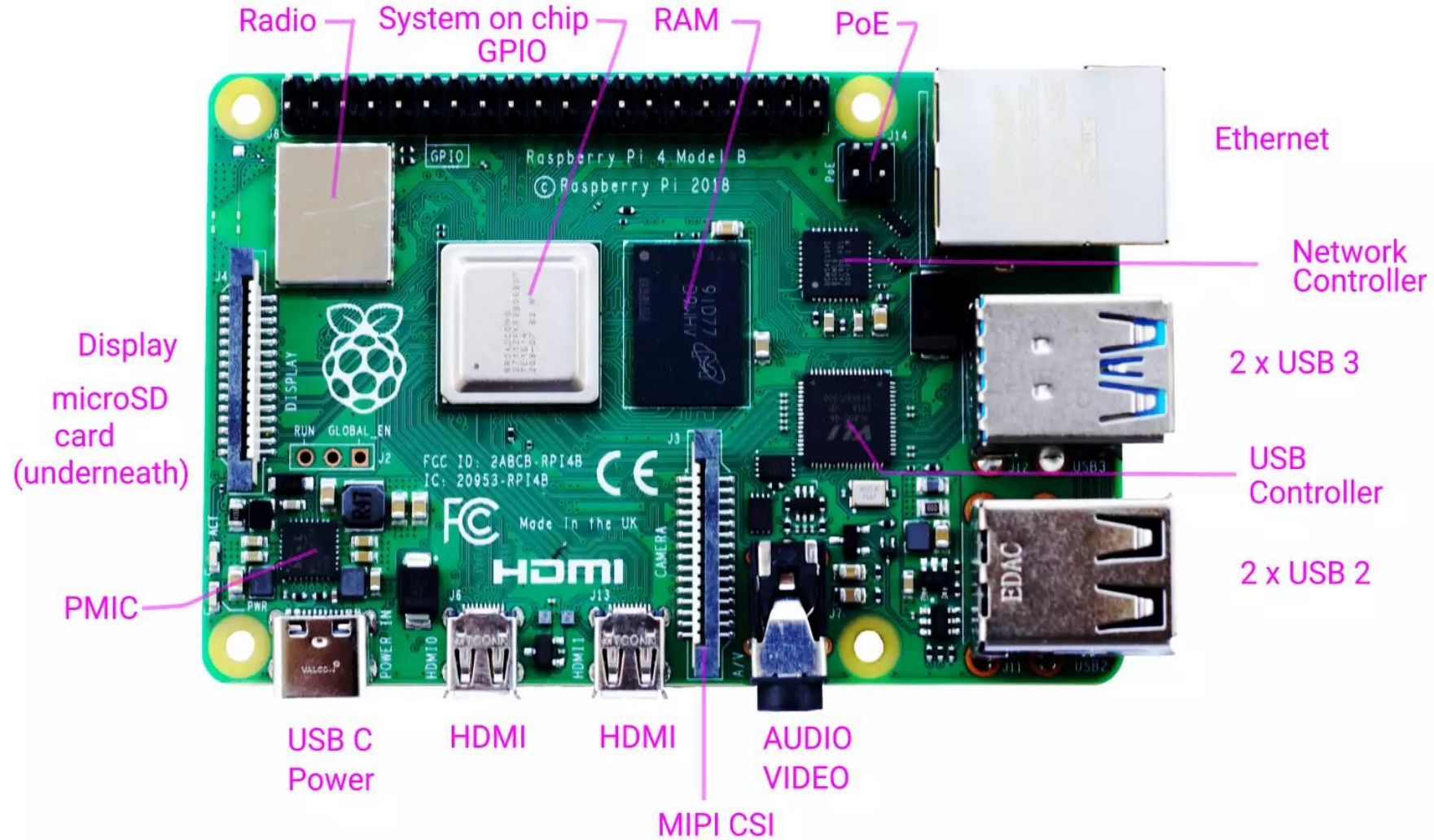
Unit- 5

Raspberry Pi

Raspberry Pi 4

Your tiny, dual-display,
desktop PC replacement.





The Raspberry Pi 4 Model B

system-on-chip, (Soc).

- This is the Broadcom BCM2711 SoC, which uses the 64-bit ARM Cortex-A72 quad-core processor as the central processing unit (CPU).
- BCM2711 also has the 32-bit Video Core VI 3D multimedia processor, or the graphics processing unit (GPU).

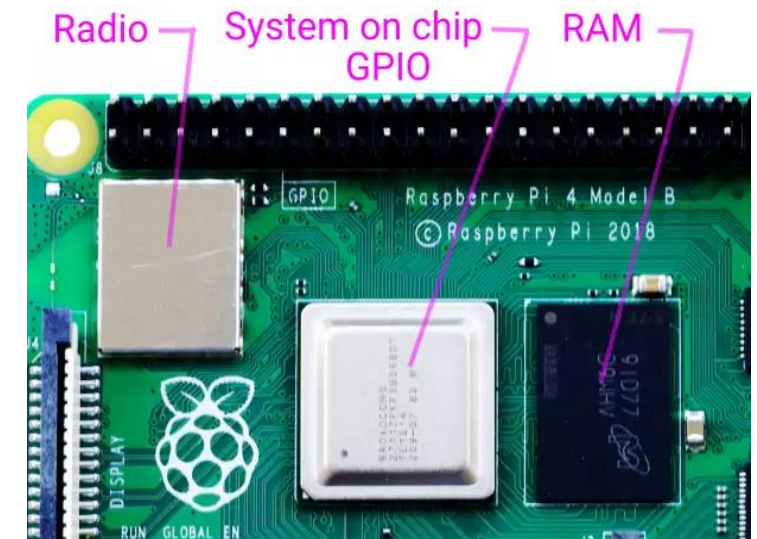


Random access memory (RAM).

- Pi4B has either 1GB or 2 GB or 4GB of LPDDR4 SDRAM.

Radio.

- The radio component comprises of two technologies. It has WiFi (IEEE 802.11 b/g/n/ac) for connecting to local area networks. There is also Bluetooth 5.0 with BLE for connecting with peripherals.



Power over Ethernet (PoE).

The PoE hat is for getting power from the Ethernet connection instead of the USB-C type port.

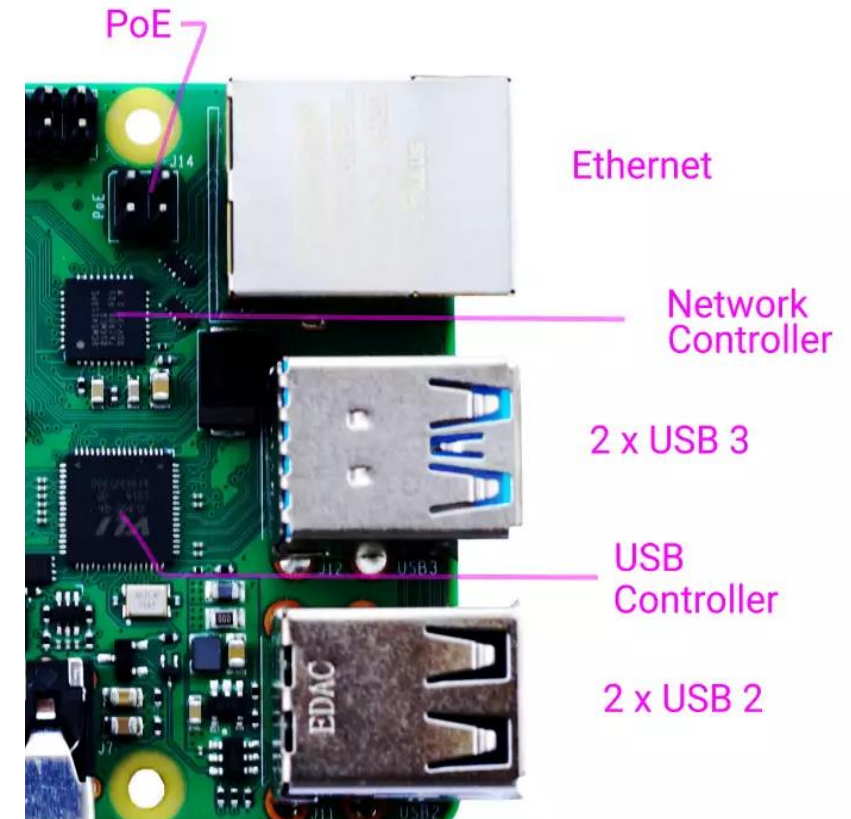
Network controller.

The network controller is for running the Ethernet interface.

USB controller. The USB controller is for managing the four USB ports.

Ethernet port.

There is an Ethernet port for linking Pi4B in a local area network



Power management integrated circuit (PMIC).

The PMIC is for taking the power from the USB C port and providing it to Pi4B.

Camera serial interface (CSI).

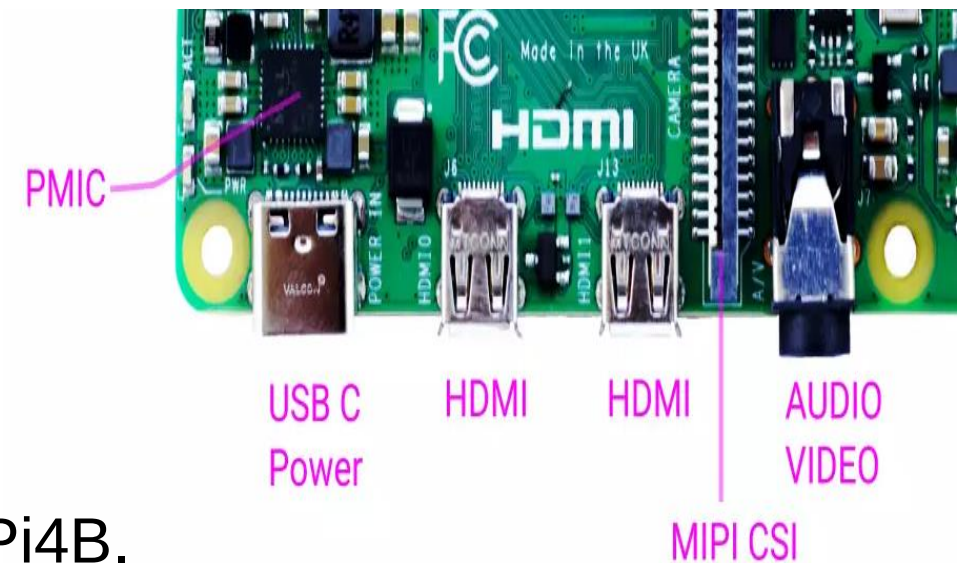
Pi4B has a MIPI Camera Serial Interface (CSI) connector. A Raspberry Pi camera module can be connected to this.

HDMI ports.

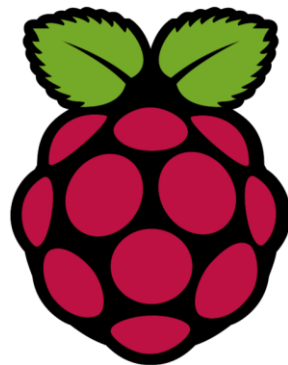
There are two micro High Definition Multimedia Interface (micro HDMI) ports.

USB Type - C power port.

The USB Type C port is for supplying power to Pi4B.



Getting Started with Raspberry Pi



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Requirements – (Hardware Components)

- Raspberry Pi 4B
- 32 GB microSD memory card
- SD Card reader (preferred USB card reader)
- Micro HDMI to HDMI cable*
- GrovePi+ kit*
- HDMI supported TV or Monitor for Display
- USB Type C power Adaptor



Install Raspberry Pi OS using Raspberry Pi Imager

Raspberry Pi Imager is the quick and easy way to install Raspberry Pi OS and other operating systems to a microSD card, ready to use with your Raspberry Pi.

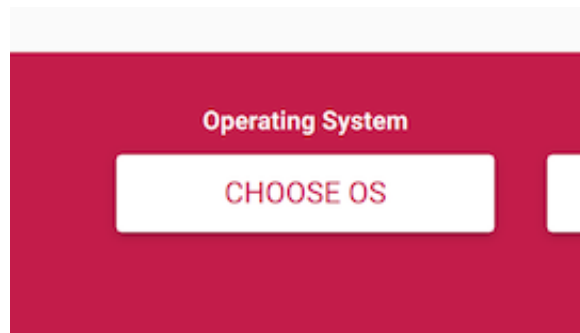
Download and install Raspberry Pi Imager to a computer with an SD card reader. Put the SD card you'll use with your Raspberry Pi into the reader and run Raspberry Pi Imager.



https://downloads.raspberrypi.org/imager/imager_latest.exe

Install Raspberry Pi OS using Raspberry Pi Imager

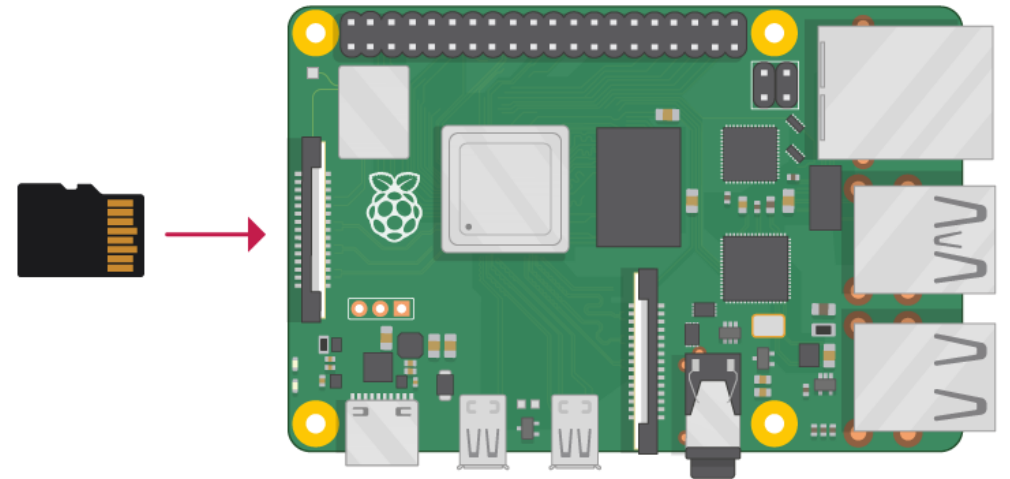
Open Raspberry Pi Imager and select OS – **Raspberry Pi OS (Legacy)**



Operating System		X
	Released: 2022-09-22 Online - 0.3 GB download	
	Raspberry Pi OS (Legacy) A port of Debian Buster with security updates and desktop environment Released: 2022-09-22 Online - 0.7 GB download	
	Raspberry Pi OS Lite (Legacy) A port of Debian Buster with security updates and no desktop environment Released: 2022-09-22	

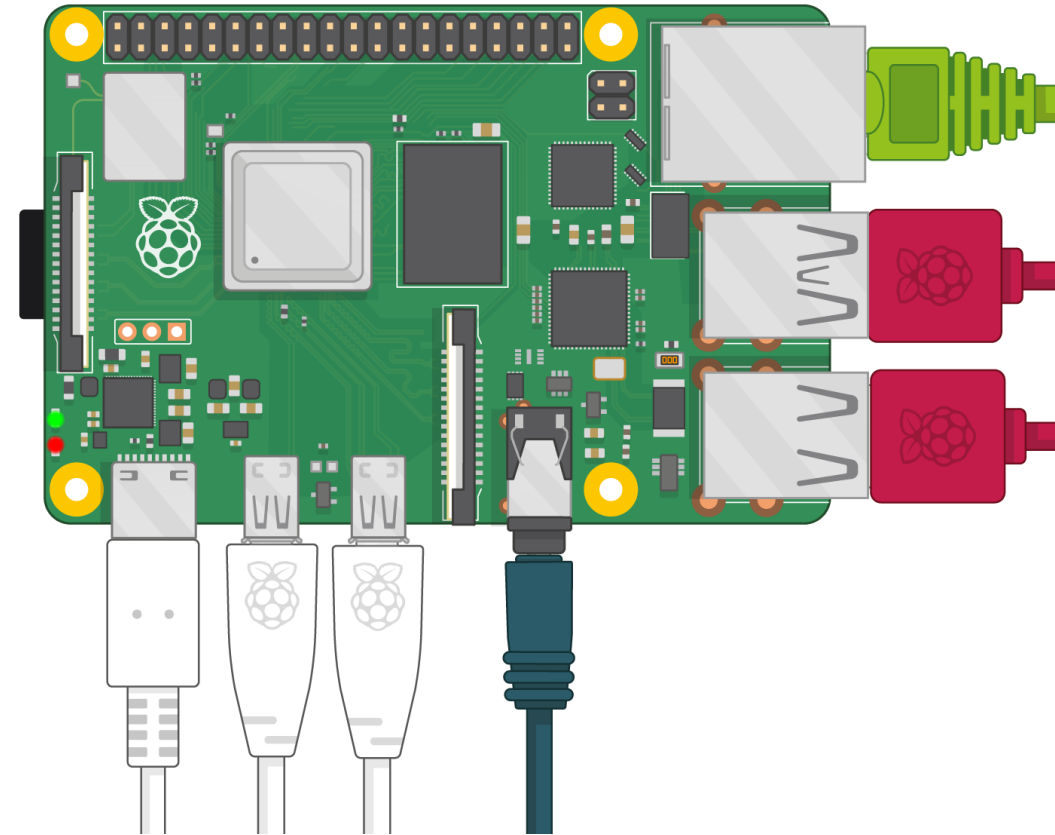
Insert SDCard

- After select OS, Select Storage device as SDCard.
- click on **Write** start writing OS to SDCard.
- Remove SDCard from Card Reader and insert into Raspberry Pi



Complete Setup

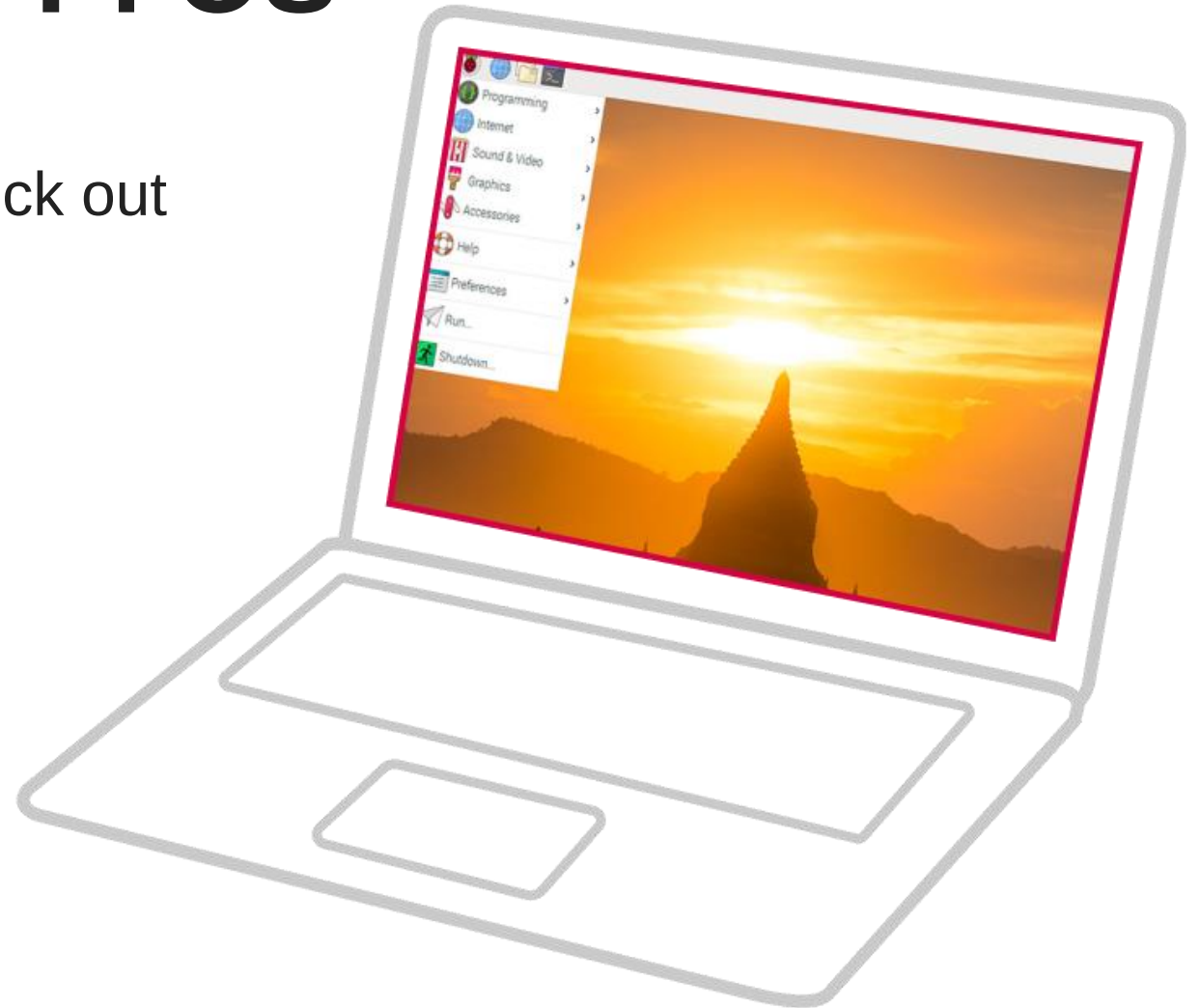
- Connect all necessary hardware to raspberry Pi as Shown in Fig.
- Connect Mouse, Keyboard to USB port
- Connect LAN cable to Ethernet port (Optional)
- Connect Display using HDMI cable
- Connect Power Supply Using Type C power Adaptor



Display Raspberry Pi OS

- After Successfully connection check out Raspberry OS on Display

Raspberry Pi Ready to Use



Welcome to Raspberry Pi Let's Explore the world of IoT